

LABORATORY 2

DIGITAL LOGIC COMPONENTS

OBJECTIVES

No.	Topics	Requirements
1	Design CMOS Logic Gates.	Implement CMOS-based NAND2, NOR2, and EXOR2 gates.
2	Implement Basic CMOS Combinational Components	Explore the design and functionality of a 2-to-1 multiplexer.
3	Implement Simple Sequential Storage Elements	Investigate the operation of a single positive-edge-triggered Modified TSPC D flip-flop.
4	[OPTIONAL] Logic Gate modeling with VerilogA language	Perform the transient simulation for the VerilogA model.
5	[OPTIONAL] Pass Transistors, Transmission Gates, and Tristate Inverters: Operation and Analysis	Analyze the operation of pass transistors, transmission gates, and tristate inverters.

EXPERIMENT 1

Objective: Implement CMOS-based logic gates.

Requirements:

- Complete the truth table, schematic, and symbol for each component.
- Run DC analysis and transient simulation.
- Create layouts for each logic gate, then show DRC confirmation and corresponding schematic with proof of LVS.

Instructions:

➤ In general, students should present their work in four parts: schematic, DC analysis, transient simulation, and layout.

a. Schematic:

The truth table of an inverter:

INPUT	OUTPUT
A	Y
0	1
1	0

Table 1 The truth table of an inverter.

The schematic of a CMOS inverter (please show us how it works):

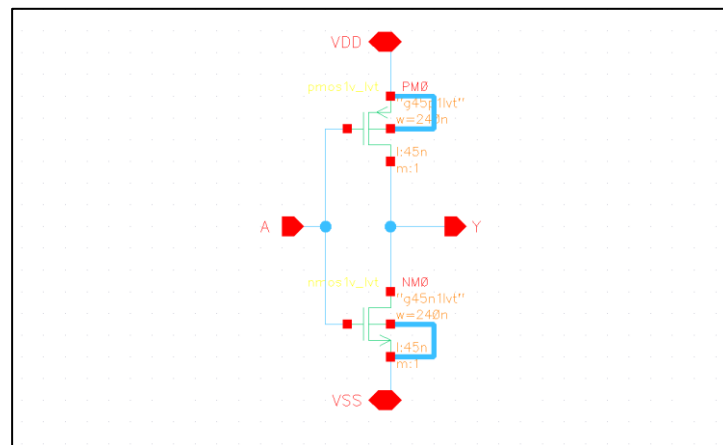


Figure 1 The schematic of a CMOS inverter.

The symbol of an inverter:

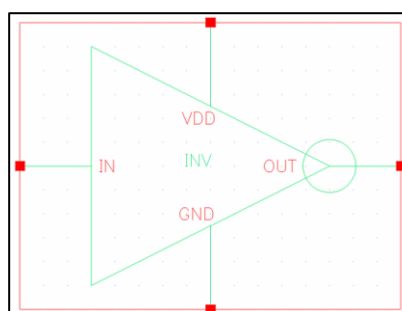


Figure 2 Symbol of INV

b. DC Analysis simulation:

Use **ADE-L** to simulate the DC response of an inverter (INV) gate. Apply an input signal as a ramp voltage ranging from 0V to 1V or perform a voltage sweep from 0V to 1V. Observe the corresponding output response. The circuit parameters are configured as follows:

Parameters	Value
V_{dd}	1V
C_{load}	1fF
V_{in}	[0, 1]V

Table 2 Parameters in DC analysis

During the simulation process, students record the following two results: output voltage values at various values of V_{in} . With a 0.1V step, plot the curve of V_{out} . Besides, *students can decrease the number of steps. V_{in} to draw a curve of V_{out} More precise.*

V_{in} (V)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
V_{out} (V)										

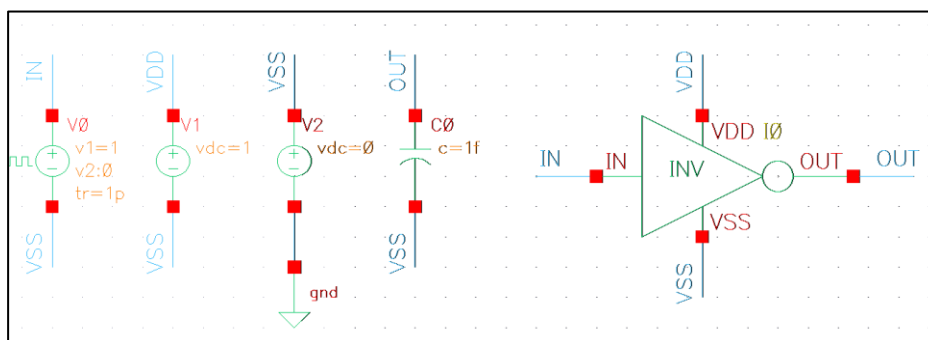
Table 3 The output voltage values at various values of V_{in} With a 0.1V step.**c. Transient simulation:**

Use ADE-Explorer to perform a time-domain simulation to verify operation in accordance with the INV gate truth table, represented as an output waveform. To conduct this simulation, assemble a testbench circuit consisting of a pulse source (vpulse), an output capacitor, and provide power to the circuit (vdc). The circuit parameters are configured as follows:

Parameters	Value	Note
V_{dd}	1V	
C_{load}	1fF	
Voltage 1	0V	Parameters for vpulse
Voltage 2	1V	
Rise time	1ps	
Fall time	1ps	
Delay	0ns	
Pulse width	1ns	
Period	2ns	

Table 4 Parameters for transient simulation.

Testbench circuit for INV

**Figure 3** The testbench of a CMOS inverter.

During the simulation, students record the input and output waveforms. Then, students measure some timing measurements.

Parameters	Result
t_{rise} – Rising time (10% – 90%)	
t_{fall} – Falling time (90% – 10%)	
t_{pdr} – Rising propagation delay (90% – 50%)	
t_{pdf} – Falling propagation delay (10% – 50%)	
t_{pd} – Average propagation delay (50% - 50%)	
Dynamic power	
Static power	

Table 5 Measurement requirements for an inverter.

d. Layout of INV:

Use the layout XL to create a logic gate layout. You need to show layout DRC and LVS violations when you run these for the first time, and how your group fixes them. You need to specify every step when you do the layout of a logic gate.

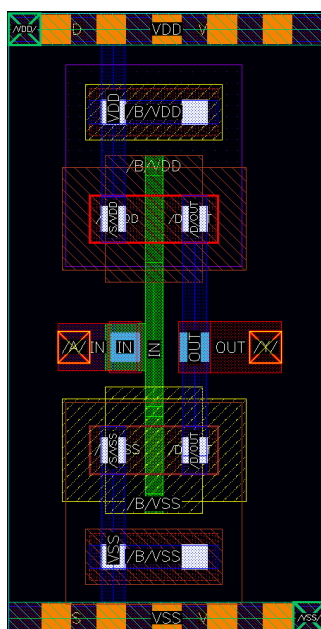


Figure 4 The complete layout of a CMOS inverter *with 1 finger*

Check: Students must show these results in the report.

➤ The sizing for each NMOS and PMOS (width/length/finger = 120n/45n/2). This size will be applied to every device in this lab. Please show these sizing parameters by capturing your testbench with these devices.

➤ The waveform to prove the circuit works correctly. Besides, students apply these parameters in **vpulse** and then simulate the design.

	In1	In2
Voltage 1	0	0
Voltage 2	1	1
Period	4n	2n

Delay time	0.65n	0.8n
Rise time	1p	1p
Fall time	1p	1p
Pulse width	2n	1n

Table 6 Testbench for NAND2, NOR2, and EX-OR2.

➤ Using the simulation result above, students measure some parameters shown in **Table 7** for each logic gate.

Parameters	Result
t_{rise} – Rising time (10% – 90%)	
t_{fall} – Falling time (90% – 10%)	
t_{pdr} – Rising propagation delay (90% – 50%)	
t_{pdf} – Falling propagation delay (10% – 50%)	
t_{pd} – Average propagation delay (50% - 50%)	
Power consumption	

Table 7 Requirements for measuring NAND2, NOR2, and EX-OR2.

- Requirement for layout:
- Input pin layer: Metal2.
 - Output pin layer: Metal2.
 - Power pin layer: Metal3.
 - Clean LVS and DRC rules.
 - Minimize the layout area.

(Hint: Students can use built-in functions in the Calculator and then reuse them for other designs)

EXPERIMENT 2

Objective: Implement Basic CMOS Combinational Components.

Requirements:

- Complete the truth table, schematic, and symbol for a 2-to-1 channel multiplexer using a compound gate.
- Run DC analysis and transient simulation.
- Create layouts for each logic gate, then show DRC confirmation and the corresponding schematic with proof of LVS.

Remind:

➤ Multiplexers are the key components in CMOS memory elements and data manipulation structures. A multiplexer selects the output from among several inputs based on a specific signal. First, students complete the truth table and give the logic function of a 2-to-1 channel multiplexer.

➤ This section presents some basic topologies of 2-to-1 MUX, such as transmission gate, compound gate, and tristate inverter.

- Two transmission gates can be tied together to form a compact 2-input multiplexer. However, the transmission gates produce a non-restoring multiplexer, and another issue is charge sharing from the output when S toggles from high to low.
- A restoring, inverting multiplexer: compound gate or gang together two tristate inverters.
- In practice, both inverting and noninverting multiplexers are simply called multiplexers or muxes.

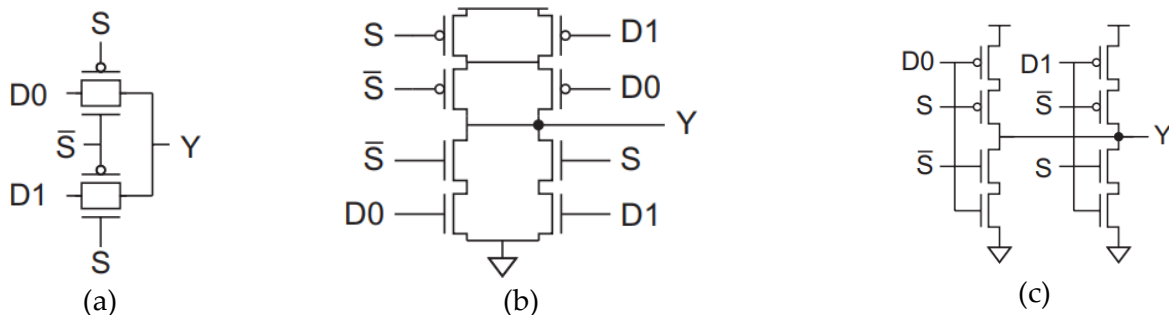


Figure 5 MUX 2-to-1 using (a) transmission gate, (b) compound gate, (c) tristate inverter. Source: from [1]

➤ Besides, the unit MUX can be composed of various types of logic gates. The detailed circuitry and pros and cons of each logic gate are summarized in the figure below:

Type	Static CMOS	Transmission Gate	Domino Logic
Scheme			
Pros	Low Power	Compact Size	Fast Operation
Cons	Large Area	No Self Driving	High Power

Figure 6 Various types of unit MUX. Source: from [2]

Check: Students must show these results in the report.

➤ The waveform to prove the 2-to-1 channel multiplexer using *compound gates* works correctly, and the measured parameters are shown in **Table 8**.

Parameters	Result
t_{rise} – Rising time (10% – 90%)	
t_{fall} – Falling time (90% – 10%)	
t_{pdr} – Rising propagation delay (90% – 50%)	
t_{pdf} – Falling propagation delay (10% – 50%)	
t_{pd} – Average propagation delay (50% - 50%)	
Power consumption	

Table 8 Measurement results of MUX 2-to-1

➤ Complete the layout of your schematic, which must pass DRC and LVS. The requirements for the layout are the same as those in experiment 1.

EXPERIMENT 3

Objective: Implement simple storage elements - single-positive-edge-triggered Modified TSPC D flip-flop.

Requirements:

- Demonstrate latch and flip-flop by studying a D latch and a D flip-flop using transmission gates.
- Complete the truth table, schematic, and symbol for each component, then run the transient simulation.
- Known definitions and how to measure setup time, propagation, and hold time.
- Find the maximum clock frequency of this element.

Instructions:

➤ Building on the previously developed combinational components, sequential circuits such as latches and flip-flops can now be designed. Figure 8(a) depicts a D latch implemented using a 2-input multiplexer, which itself can be realized using a pair of transmission gates, as shown in Figure 8(b).

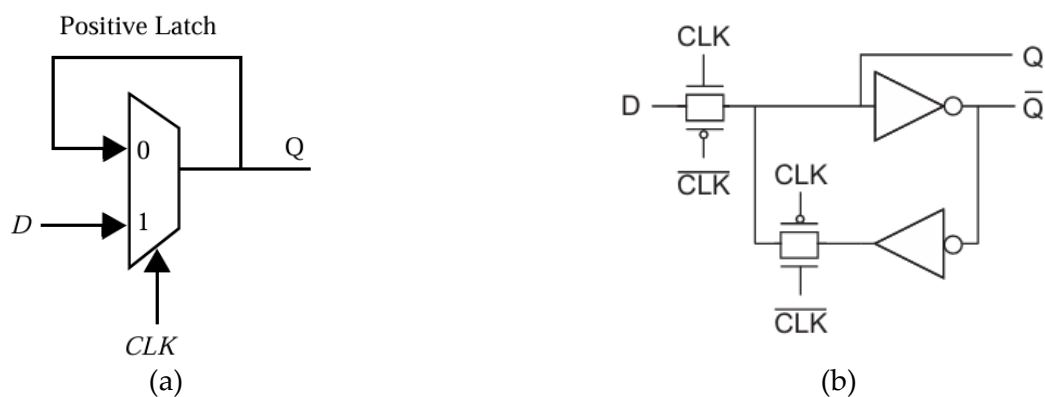
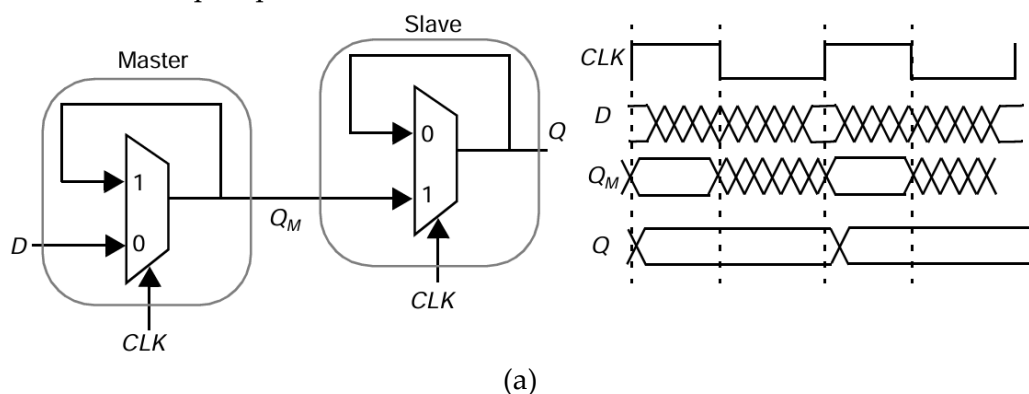


Figure 7 (a) Positive D-latch based on a multiplexer. Source: from [3] ; (b) CMOS positive-level-sensitive D latch. Source: from [1]

➤ By combining two level-sensitive latches—one negative-sensitive in the first stage and one positive-sensitive in the second stage—a positive-edge-triggered flip-flop is formed. As shown in **Figure 9(a)**, the first latch, known as the master, captures the input data, while the second latch, called the slave, transfers the data on the clock's rising edge. This device is commonly referred to as a D flip-flop, D register, or master-slave flip-flop.



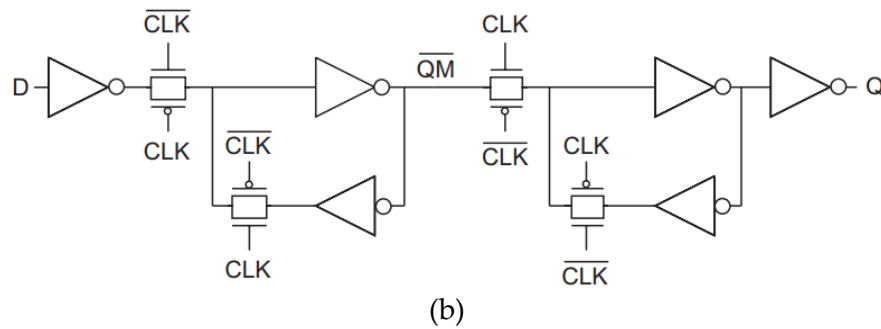


Figure 8 (a) Positive edge-triggered register based on a master-slave configuration. Source: from [3](b) Schematic of master-slave positive edge-triggered register using multiplexers. Source: from [1]

➤ The negative edge-triggered flip-flop is based on a master-slave architecture and is designed to be insensitive to clock overlap. This design, known as the Clocked CMOS (C2MOS) flip-flop, incorporates dynamic latches for efficient operation. As shown in **Figure 10(a)**, the C2MOS flip-flop consists of dynamic latch topologies, where the data signal is processed by PMOS transistors at the top and NMOS transistors at the bottom. This configuration functions as a clocked gate, integrating both logic and latch operations.

- When the clock signal (**CLK**) is HIGH (1), the input is inverted, and the output is generated. Conversely, when **CLK** is LOW (0), the output enters a high-impedance state, preventing data propagation. **Figure 10(b)** illustrates the schematic of a C2MOS D flip-flop, which consists of two cascaded C2MOS latches, ensuring reliable data storage and transfer.

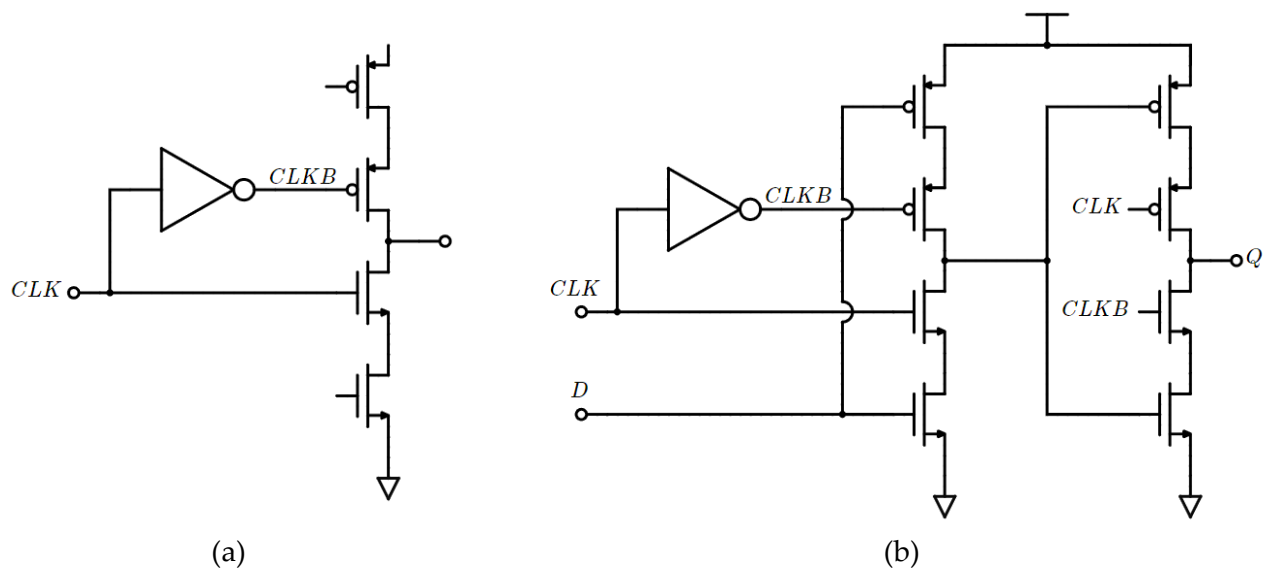


Figure 9 (a) A clocked CMOS latch; (b) The schematic of C2MOS negative-edge-triggered DFF. Source:

Adapted from [3]

➤ True Single-Phase Clock (TSPC) [4] is a common dynamic Flip-Flop design that offers higher speed, smaller area, and lower power dissipation than other Flip-Flop architectures. Due to the requirement for sharp clock edge transitions and the stronger pull-up provided by PMOS devices, the design operates as a positive-edge-triggered Flip-Flop.

- When the clock signal CLK is 0, the input is isolated from the output, as the third stage's input is precharged to HIGH, allowing Q to retain its previous value. When CLK is 1, the

third stage's input remains unaffected. Therefore, as long as CLK is stable at either HIGH or LOW, the input remains isolated from the output. However, during a LOW-to-HIGH clock transition, QB latches the complement of the input while Q propagates the input to the output.

- Analyzing the behavior of node B reveals that whenever there is a discharge path to ground, the third stage's input is always precharged to HIGH when CLK is LOW and resets to LOW when CLK is HIGH. Consequently, if the input D remains at a stable LOW for an extended period, the third stage undergoes continuous toggling. This unnecessary switching not only contributes to excessive power consumption but also introduces noise at the output node Q, causing spurious glitches during each LOW-to-HIGH clock transition.
- A simple yet effective technique to mitigate this issue is to incorporate a PMOS transistor, which prevents the precharging phase from occurring without affecting the global operation of the Flip-Flop (Figure 11(b)).

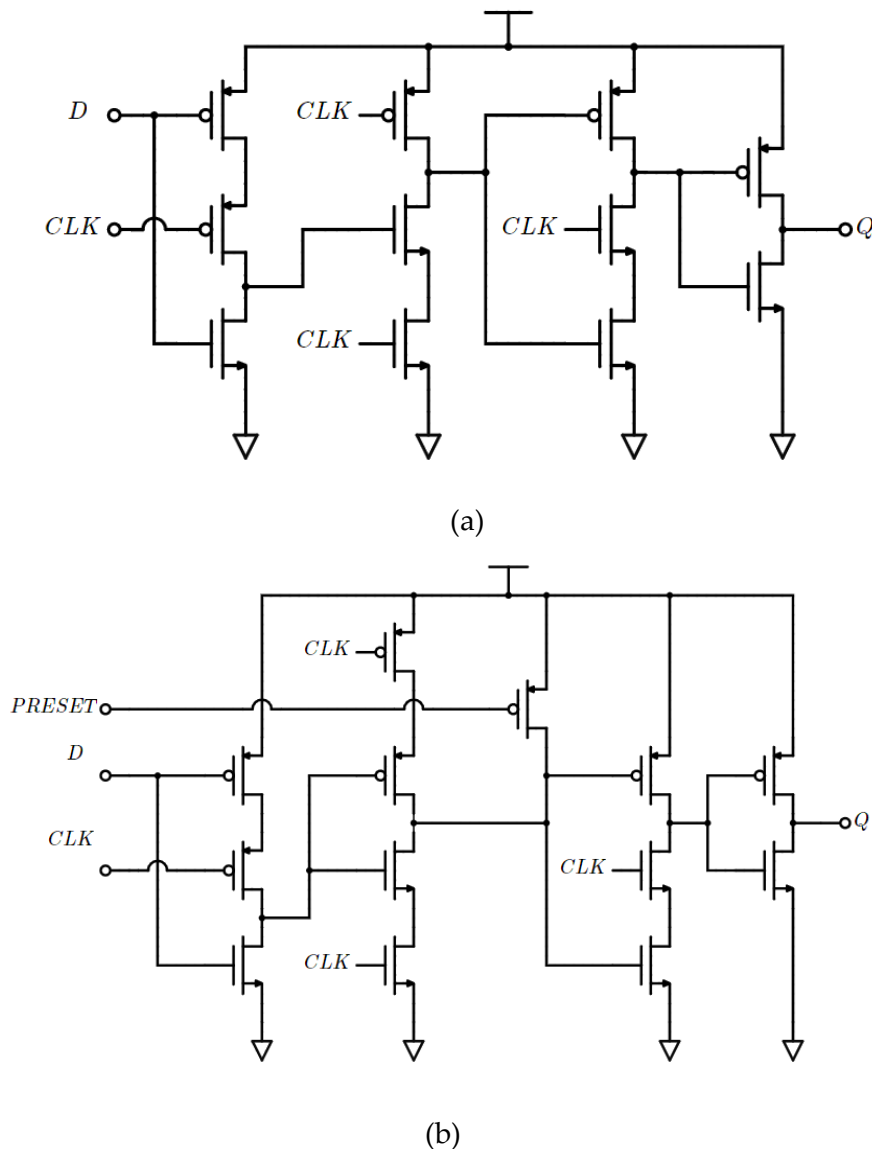


Figure 10 The schematic of (a) negative-edge-triggered TSPC DFF, (b) positive-edge-triggered Modified TSPC DFF with Preset.

➤ The Dual Edge-Triggered Flip-Flop (DETFF) samples data on both the rising and falling edges of the clock signal. Its design is based on multiplexers and consists of two distinct data paths.

- The upper path features a Single Edge-Triggered (SET) flip-flop implemented using PMOS pass transistors, which operate on the falling edge of the clock. Similarly, the lower path consists of an SET flip-flop but utilizes NMOS pass transistors, functioning on the rising edge of the clock.
- Both data paths are connected to feedback loops formed by two cascaded inverters. Additionally, each path includes a transistor (NMOS for the upper path, PMOS for the lower path) positioned after the loop. This transistor acts as a switch, controlled by the inverted clock signal.

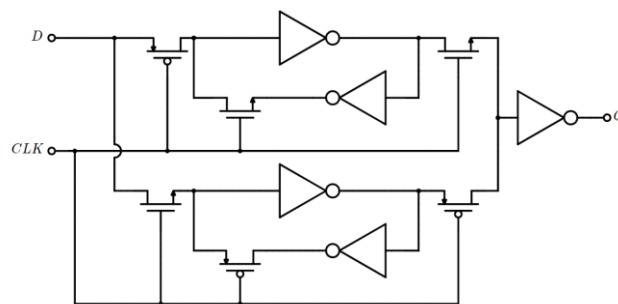


Figure 11 Schematic of a D Flip-Flop dual-edge triggered.

➤ In this experiment, students use Cadence Virtuoso to measure the setup and hold time of a D flip-flop. Recalling that the setup and hold time is the minimum time before and after the rising-edge clock, the input signal must remain constant to store the signal and to generate a stable output, respectively.

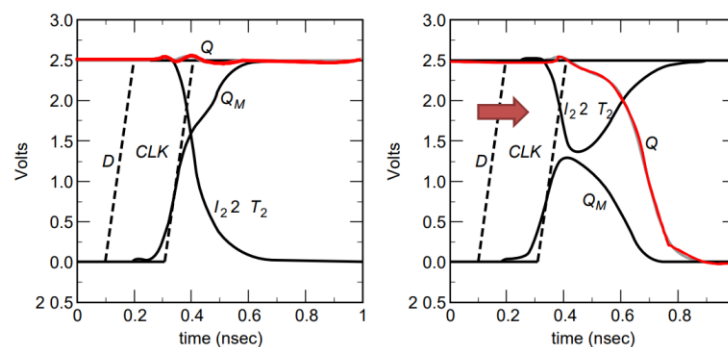


Figure 12 Overview of measurement setup and hold time method. Source: from [3]

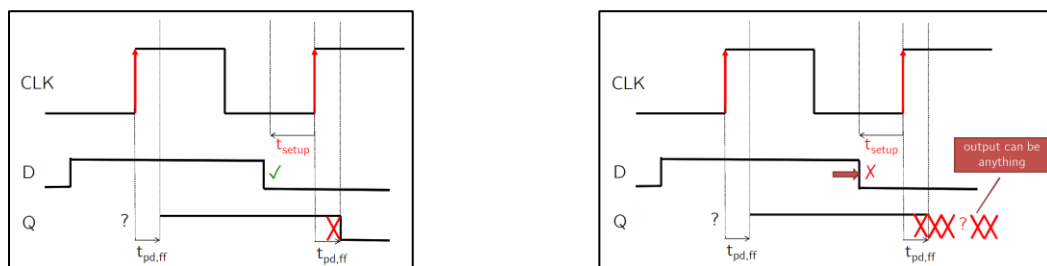


Figure 13 Waveform describing the measurement setup time method. Source: from [4]

➤ Students can measure setup and hold time using this method: *Shift the input D closer to the rising CLK signal until the Q output is incorrect.*

➤ In addition, the table below defines the delays and timing constraints of the combinational logic and sequencing elements.

Term	Name
t_{pd}	Logic Propagation Delay
t_{cd}	Logic Contamination Delay
t_{pcq}	Flop Clock-to-Q Propagation Delay
t_{ccq}	Flop Clock-to-Q Contamination Delay
t_{setup}	Flop Setup Time
t_{hold}	Flop Hold Time

Table 9 Sequencing element timing notation.

Figure 15(a) shows the response of the combinational logic to the input A changing from one arbitrary value to another. **Figure 15(b)** shows the response of the flip-flop.

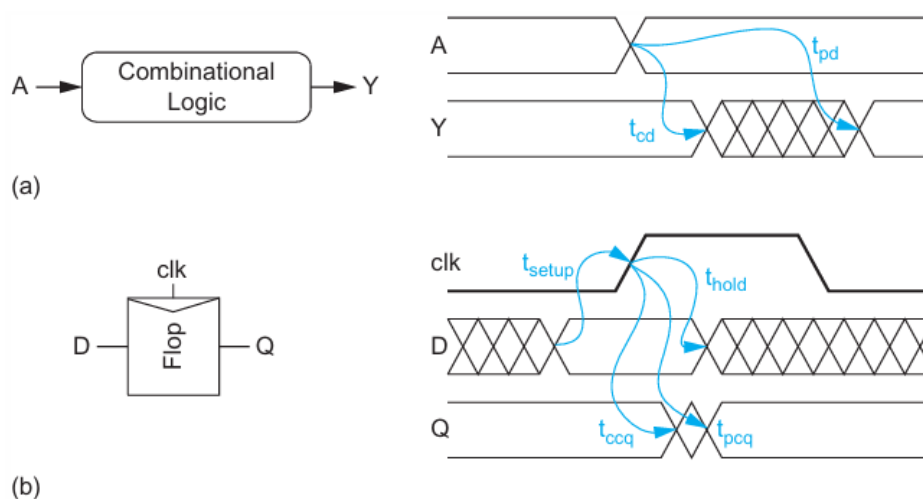


Figure 14 Timing diagrams. Source: from [1]

Check: You must show in your report.

➤ Simulate the *positive-edge-triggered* TSPC D flip-flop without the Preset signal (reference in **Figure 11(b)**). In addition, measure some timing parameters and power consumption.

Parameters	Result
t_{setup} – Flop setup time	
t_{hold} – Flop hold time	
t_{pcqr} – Flop Clock-to-Q Propagation Delay (Low to High)	
t_{pcqf} – Flop Clock-to-Q Propagation Delay (High to Low)	
t_{pcq} – Flop Clock-to-Q Propagation Delay (average)	
Power consumption	

Table 10 Measurement results of positive-edge-triggered TSPC D flip-flop.

EXPERIMENT 4 [OPTIONAL]

Objective: Create and perform the simulation of logic gates using VerilogAMS Modeling.

Requirements

- Explore and create the VerilogA code for the NAND2, NOR2, and EXOR2 logic gate operations.
- Perform the simulation for the VerilogA model of the above logic gate.

Instructions:

- As a reference from LAB 0 – Custom Digital IC Design flow, you can follow to create a VerilogA code to model some ideal components (logic gate, decoder, storage elements).
- Create the testbench for these logic gates and test the **transient response** for each model.
- Create the new cell for this experiment; do not create the VerilogA view inside the experiment 1 cells.

Checks:

- Students need to create a new testbench for this experiment, report the same as experiment 1, with the same settings of the vpulse signals for the test.
- Capture the transient response and show that the rise/fall times are the same as the VerilogA code configuration

Voltage 1	0	0
Voltage 2	1	1
Period	4n	2n
Delay time	0.65n	0.8n
Rise time	1p	1p
Fall time	1p	1p
Pulse width	2n	1n

Table 11 Test the vpulse signal for NAND2, NOR2, and EX-OR2 VerilogA modeling.

- Reference source: <https://designers-guide.org/verilog-ams/index.html>

EXPERIMENT 5 [OPTIONAL]

Objective: Review the operation of the pass transistor, transmission gate, and tristate inverter.

Requirements

- Analyze the charge and discharge period of MOS transistors, then prove by simulation.
- Analyze the operations of a transmission gate, then prove by simulation.

Instructions:

- Schematic, symbol, and operation description of the pass transistor, transmission gate, and tristate inverter.

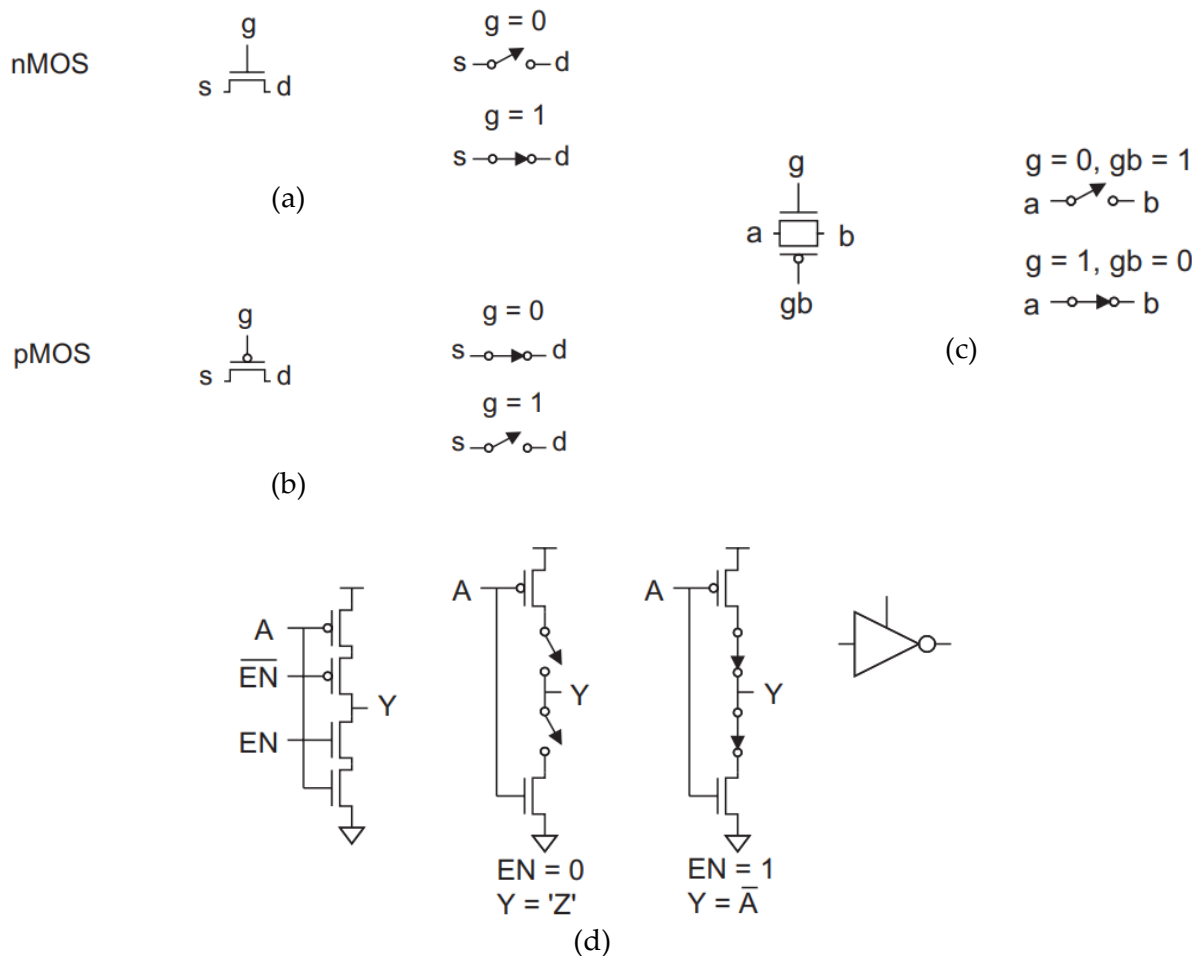


Figure 15 Schematic and operation of (a) pass transistor NMOS, (b) pass transistor PMOS, (c) transmission gate, (d) tristate inverter. Source: from [1]

- Following these schematics shown in **Figures 17** and **18**, students need to calculate and define when NMOS turns on, and determine $v_C(0^-)$ Value (shown in **Figure 17**), and vice versa, PMOS (shown in **Figure 18**).

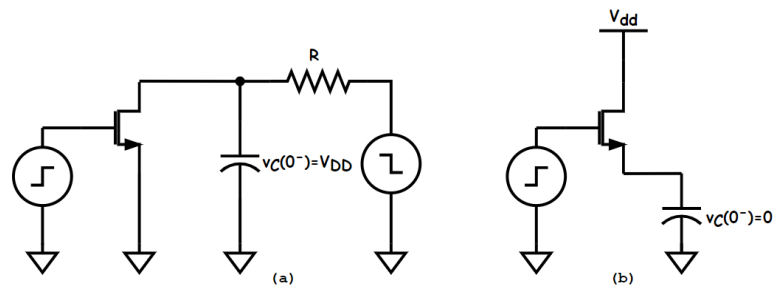


Figure 16 Testbench when (a) NMOS discharging, (b) NMOS charging.

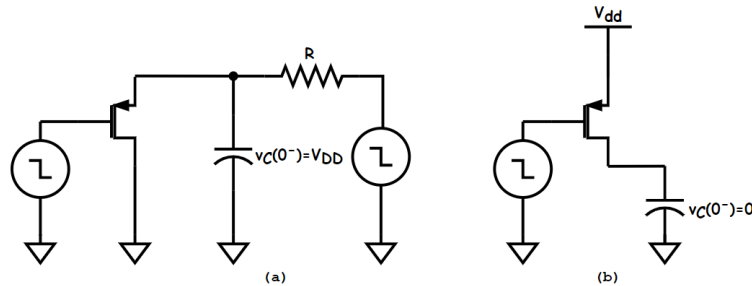


Figure 17 Testbench when (a) PMOS discharging, (b) PMOS charging.

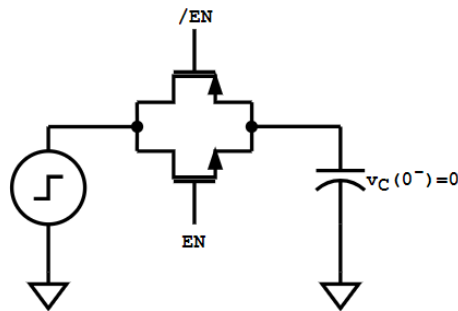


Figure 18 Schematic of transmission gate (testbench)

Check:

- Students create schematics, then simulate and show waveforms of the circuit as shown in **Figure 17** and **Figure 18**, and then re-check the calculations students perform in the instruction section.
- Students create schematics and then simulate and show waveforms of the circuit as shown in **Figure 19**.

Questions:

- According to the first check requirement, please let us know in which cases you use MOS transistors as a switch.
- The second check requirement shows us some advantages and disadvantages when transmission gates are used as a switch.
- Comment on the position of Drain and Source to Gate in MOSFET's structure. How can you demonstrate them? In your opinion, data transmitted in a pass transistor and transmission gate can only flow in one direction, cannot it? Explain your idea.

ACKNOWLEDGEMENTS

This laboratory has been developed and refined through the contributions of numerous individuals over the years. It was originally designed for the *EE3117 Digital IC Design* course at the University of Technology – VNU, HCM by *Mr. Nguyen Trung Hieu* (2015) and *Mr. Duong Quang Ho* (2018). Subsequent revisions and enhancements have been implemented for the *EE3117 Digital IC Design* course under the guidance of Dr. Linh Tran, with additional contributions from Doanh Bui, Tan-Khai Pham, and Phuc T. Nguyen-Phan.

Versions of this lab have been used in the following courses:

- EE3117 Digital IC Design (2023 – present) – the University of Technology – VNU, HCM.
- EE5154 Advanced Digital IC Design (2024 – present) – the University of Technology – VNU, HCM.

REFERENCES

- [1] N. H. E. Weste and D. M. Harris, “CMOS VLSI Design 4th Ed.” Accessed: Feb. 24, 2025. [Online]. Available: <https://pages.hmc.edu/harris/cmosvlsi/4e/index.html>
- [2] D. Chon and W. Choi, “Hardware Efficient Transposable 8T SRAM for Orthogonal Data Access,” *IEEE Access*, vol. 12, pp. 981–992, 2024, doi: 10.1109/ACCESS.2023.3347913.
- [3] J. M. Rabaey, A. Chandrakasan, and B. Nikolić, “Digital Integrated Circuits Second Edition.” Accessed: Apr. 03, 2025. [Online]. Available: <https://icbook.eecs.berkeley.edu/resources/powerpoint-slides>
- [4] “ECE 4740 Course Details – Cornell ECE Open Courseware.” Accessed: Apr. 03, 2025. [Online]. Available: <https://ocw.ece.cornell.edu/ece-4740-course-details/>

